

REMIND: Retrieval-Augmented Reconstruction With Dual Memories for Modality-Missing Object Re-Identification

Zhendong Xu¹, Zi Wang², Aihua Zheng³, Chenglong Li⁴, *Senior Member, IEEE*, and Jin Tang⁵

Abstract—To address the modality-missing object Re-Identification (Re-ID) task, a common strategy is to compensate for absent information by exploiting available modalities. However, existing reconstruction-based approaches suffer from two major limitations: 1) they often overlook modality-specific cues inherent in the missing modality; 2) they typically adopt a single-path reconstruction strategy. These issues result in incomplete representations and constrain the capacity to model complex semantic mappings across heterogeneous modalities. To address these challenges, we propose REMIND, a novel framework for modality-missing object Re-Identification, namely REtrieval-Augmented Reconstruction With Dual Memories. Specifically, we design a Dual Memory Construction module that, guided by information-theoretic insights, extracts modality-specific and modality-common features through two complementary branches and stores them in dedicated memory banks. These memory banks serve as structured prior knowledge to guide the reconstruction process, ensuring that the features of missing modalities are preserved even under modality-missing conditions. In addition, we have developed a retrieval-augmented missing reconstruction module that enhances the expressiveness and robustness of the reconstruction through multi-path reconstruction and perturbation mechanisms. Adaptive fusion techniques are employed for integration, simultaneously improving the expressiveness and robustness of the reconstructed features. Through the synergy of information-theoretically motivated regularization and retrieval-enhanced reconstruction, REMIND achieves robust feature recovery and delivers highly discriminative representations for reliable modality-missing

Re-ID. Extensive experiments on several multi-modal object Re-ID benchmarks demonstrate the effectiveness and superiority of REMIND under various missing modality scenarios. The code is publicly available at: <https://github.com/skye-1201/REMIND>

Index Terms—Modality missing, object re-identification, memory bank, feature reconstruction.

I. INTRODUCTION

MULTI-MODAL object Re-ID aims to identify the same object across different camera views by leveraging complementary information from multiple spectra. With the release of several paired multi-modal datasets [1], [2], [3], [4], this task has achieved notable progress [5], [6], [7], [8], [9], [10]. However, in real-world scenarios, missing modalities are common due to sensor failures or transmission errors, leading to performance degradation in fusion-based or interaction-based methods [2], [11], [12], [13], [14].

To mitigate performance degradation under modality-missing conditions, existing methods typically reconstruct the absent features from the available ones through transformation modules. However, as illustrated in Fig. 1 (a), these approaches inherently suffer from two limitations. First, the Venn diagram reveals that different modalities share a certain portion of common semantics while also containing distinct modality-specific cues. Current methods mainly rely on the common information to recover the missing modality, since this part is more easily transferable across modalities. Yet, by focusing solely on robust common semantics, they inevitably overlook the non-overlapping regions that encode modality-specific discriminative information. As a result, the reconstructed features fail to completely capture the unique cues embedded in the missing branch, leading to insufficient discriminative power [15], [16], [17]. In addition, most existing works adopt a single-path reconstruction strategy, in which available features are transformed via a fixed mapping to approximate the missing modality. The heatmap in Fig. 1 (a) demonstrates that such single-path reconstruction produces representations with only partial discriminative activations (red boxes), while the remaining identity-relevant regions are only weakly activated or even missing (green boxes). This limitation arises because a single mapping cannot adequately model the complex and diverse semantic interactions that occur across heterogeneous modalities, nor can it capture their hierarchical dependencies. Consequently, the reconstructed features remain coarse and structurally incomplete, restricting the model's capacity for

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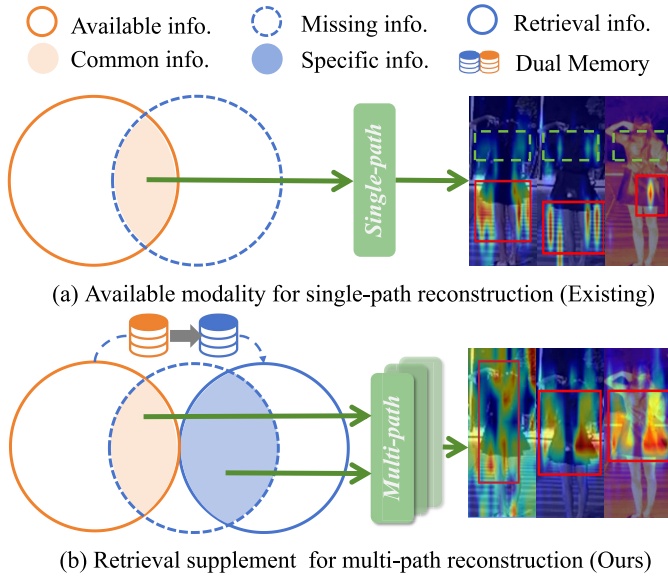


Fig. 1. Comparison with existing reconstruction methods for modality-missing scenarios. (a) Existing reconstructions: single-path reconstruction for limited information. (b) Proposed: introducing modality-specific information and multi-path reconstruction from more valuable information.

fine-grained alignment and complementary feature modeling [18], [19], [20].

To address these challenges, we propose the Retrieval-Augmented Reconstruction with Dual Memories (REMIND) framework. This framework is designed to complement the discriminative cues of missing modalities and enhance the cross-modal reasoning capability of the reconstruction module. REMIND consists of two key components: the Dual Memory Construction module, which captures both modality-specific and modality-common information, and the Retrieval-augmented Missing Reconstruction module, which aims to reconstruct robust representations by leveraging information from the retrieval.

Firstly, to capture robust modality-specific and modality-common information for missing modality completion, we propose a dual memory construction strategy: motivated by information-theoretic insights, this design separately extracts modality-specific cues and common semantic features, which are stored in their corresponding memory banks. In the modality-specific feature extraction branch, interaction with fixed textual prompts enhances the extraction of fine-grained features. To further regularize representation learning, we introduce a modality-specific regularization term motivated by the information-theoretic insights that reducing the dependency between modality-specific features and fused features helps suppress redundant common information. This loss incorporates variance-aware weights across modalities for adaptive fusion [21], [22], and imposes an orthogonality constraint between uni-modal and fused representations, effectively preserving unique modality-specific signals while removing redundant common information [23]. The modality-common feature extraction branch aggregates multi-modal inputs through a common-information enhancement branch, projecting them into a shared space to learn semantically consistent features. To promote semantic alignment

across modalities, we further introduce an information-theoretically motivated consistency regularization term. This term suppresses modality-specific noise and enforces identity consistency across modalities. Finally, inspired by prior studies on memory-based representation learning [24], [25], we construct dual memory banks to encode modality-specific and modality-common semantics, serving as structured priors for subsequent reconstruction and retrieval. The modality-specific memory bank stores modality-unique information to support missing modality reconstruction, while the modality-common memory bank acts as a buffer for common features, alleviating cross-modal inconsistency during retrieval.

Second, existing feature transformation methods are typically designed independently, making it difficult to collaboratively recover discriminative information from multiple perspectives [12], [26], [27]. Moreover, retrieval-based completion approaches often suffer from cross-modal inconsistency when directly retrieving information [28], [29], [30]. To address these challenges, we propose a Retrieval-augmented Missing Reconstruction (RMR) method. This method leverages modality-common features as semantic anchors to retrieve identity-consistent entries from a common memory bank, which serve as reliable priors to guide subsequent retrieval from modality-specific memory banks. Such a hierarchical retrieval strategy effectively decouples and complements modality-common and modality-specific semantics, ensuring that the reconstructed features remain both identity-consistent and semantically complete. Subsequently, the retrieved features and the available modality features are jointly fed into a multi-path reconstruction module. In this module, each reconstruction path incorporates noise perturbation to simulate uncertainty in feature space, thereby enhancing robustness and generalization capability. Finally, a threshold-aware fusion mechanism adaptively aggregates the multi-path reconstruction candidates to generate a semantically rich and reliable representation of the missing modality. This “*identify first, then retrieve*” strategy alleviates the inherent cross-modal inconsistency in direct retrieval, while the perturbation-based multi-path reconstruction further enhances the model’s capacity to capture complex semantics and perceive feature uncertainty.

We evaluate our framework under both missing modality and complete modality settings on four multi-modal object Re-ID datasets: RGBNT201 [2], RGBNT100 [1], WMVEID863 [4], and MSVR310 [3]. Extensive experimental results clearly demonstrate the effectiveness and generalization of our proposed approach.

This paper makes the following contributions:

- We propose the REMIND framework to address the modality-missing challenge in multi-modal object Re-ID. REMIND effectively supplements discriminative cues from the missing modality and enhances the modeling capacity of the reconstruction module.
- To address the lack of modality-specific cues in reconstructed features, we propose the Dual Memory Construction (DMC) module to explicitly disentangle and store modality-specific and common information.
- To overcome the limitation of single-path reconstruction, we design the Retrieval-augmented Missing

Reconstruction (RMR) module that generates diverse reconstructions and adaptively fuses them for robustness.

- Extensive experiments on benchmark multi-modal Re-ID datasets under both complete and modality-missing settings demonstrate the superior effectiveness and robustness of the proposed REMIND framework.

II. RELATED WORK

A. Multi-Modal Object Re-ID

Multi-modal object Re-ID has attracted increasing attention due to its robustness under complex scene variations. Zheng et al. [5] propose the IEEE framework to learn stable multi-modal person representations. Li et al. [1] introduce a multi-modal vehicle Re-ID baseline, HAMNet, to effectively fuse features from different modalities. To address the discrepancies across samples and modalities, Zheng et al. [3] propose CCNet, which employs distance constraints to reduce inconsistency. Pan et al. [31] adopt a Transformer-based architecture that fuses randomly cropped multi-modal images at the feature level. Wang et al. [32] design a novel training strategy, HTT, which fine-tunes the model before inference to improve generalization. Additionally, Zhang et al. [33] propose a new framework, EDITOR, to mitigate the challenges introduced by complex Re-ID backgrounds. Wang et al. [34] propose MambaPro, a multi-modal Re-ID framework designed to model long-range dependencies across modalities. Liu et al. [35] propose Signal, which performs selective token interaction and global-local alignment to suppress background interference and improve multi-modal consistency. Xu et al. [36] propose STMI, which uses segmentation-guided token modulation to highlight identity-relevant regions and cross-modal hypergraph interaction to model high-order modality correlations. Li et al. [37] leverage learnable textual prompts to improve representation quality by mining fine-grained information. Furthermore, Wang et al. [9] incorporate multi-modal textual cues to enrich semantic information. However, most of these approaches assume completely available modalities and suffer performance degradation in real-world scenarios where missing modality data is common.

B. Modality-Missing Object Re-ID

Recently, growing attention has been paid to modality-missing settings in multi-modal Re-ID. Existing methods can be categorized into: (1) Generation-based approaches, which explicitly synthesize features of the missing modality [16], [38], [39]; (2) Joint learning methods, which aim to mitigate the effect of missing modalities by enhancing the discriminative capacity of available inputs through robust common representations [40], [41], [42], [43]. Some recent works have begun addressing the missing modality problem in multi-modal Re-ID. Zheng et al. [2] propose PFNet, which converts RGB features into NIR and TIR modalities to compensate for missing data. Considering the possibility of arbitrary modality absence, Wang et al. [12] propose a novel Re-ID network to perform recurrent multi-modal feature aggregation and complementary reconstruction. Wang et al. [13] further design a multi-expert framework to learn more robust common representations under incomplete modality conditions. Zhang et al.

[14] propose PromptMA to mitigate the modality deficiency problem by introducing prompt learning to model robust multi-modal complementary information. However, these methods often rely solely on the common features from the available modalities, failing to recover the modality-specific information of the missing branch, leading to suboptimal discriminative power in the reconstructed features. Additionally, their single-path reconstruction modules lack the capacity to model the complex and heterogeneous relationships across modalities. In contrast, REMIND tackles these challenges by combining refined memory storage with multi-branch modeling, leading to improved robustness in missing-modality scenarios.

III. METHOD

In this section, we first present the baseline framework, then elaborate on our proposed Retrieval-Augmented Reconstruction With Dual Memories method for enhancing missing modality reconstruction. REMIND consists of: (1) **Dual Memory Construction (DMC)**, which extracts modality-specific and modality-common features via textual prototype interaction and tailored supervision, storing them as semantic priors in structured memory banks; (2) **Retrieval-augmented Missing Reconstruction (RMR)**, which retrieves relevant identity-aware features based on available inputs and generates robust reconstructions using multi-path ReBlocks with weight-aware fusion.

A. The Baseline Framework

As shown in Fig. 2, REMIND employs three CLIP-based visual encoders to extract features from visible (RGB), near-infrared (NIR), and thermal infrared (TIR) images:

$$\text{input} = [I_R, I_N, I_T], \quad f_m = \text{CLIP}_m(I_m), \quad (1)$$

where $[H, W, 3]$ is the input image size, $m \in M = [R, N, T]$ and f_m denotes features from modality m . The strong cross-modal alignment ability and semantic richness of CLIP [44] make it well-suited for this task. Additionally, fixed textual prompts are extracted via the text encoder of CLIP to provide semantic guidance:

$$P_t = \text{CLIP}_t(\text{text}), \quad \text{text} = \text{"a photo of a [cls]"}, \quad (2)$$

where $[cls]$ denotes the identity class (person or vehicle).

B. Dual Memory Construction

To obtain discriminative cues from the missing modality and enhance feature robustness, we propose the Dual Memory Construction (DMC) module, which decouples modality-specific and modality-common semantics via two branches: one preserves fine-grained traits using textual prototypes and a dedicated loss, while the other learns identity-consistent features by suppressing modality noise. These features are stored in structured memory banks as semantic priors to support reliable retrieval even when some modalities are entirely absent. To further regularize the extraction of modality-specific and modality-common information, we characterize two dependency relationships from an information-theoretic perspective as:

$$I = I(f_m^{sp}; F_{\text{fuse}}) + I(f_{m_1}^{co}; f_{m_2}^{co}), \quad (3)$$

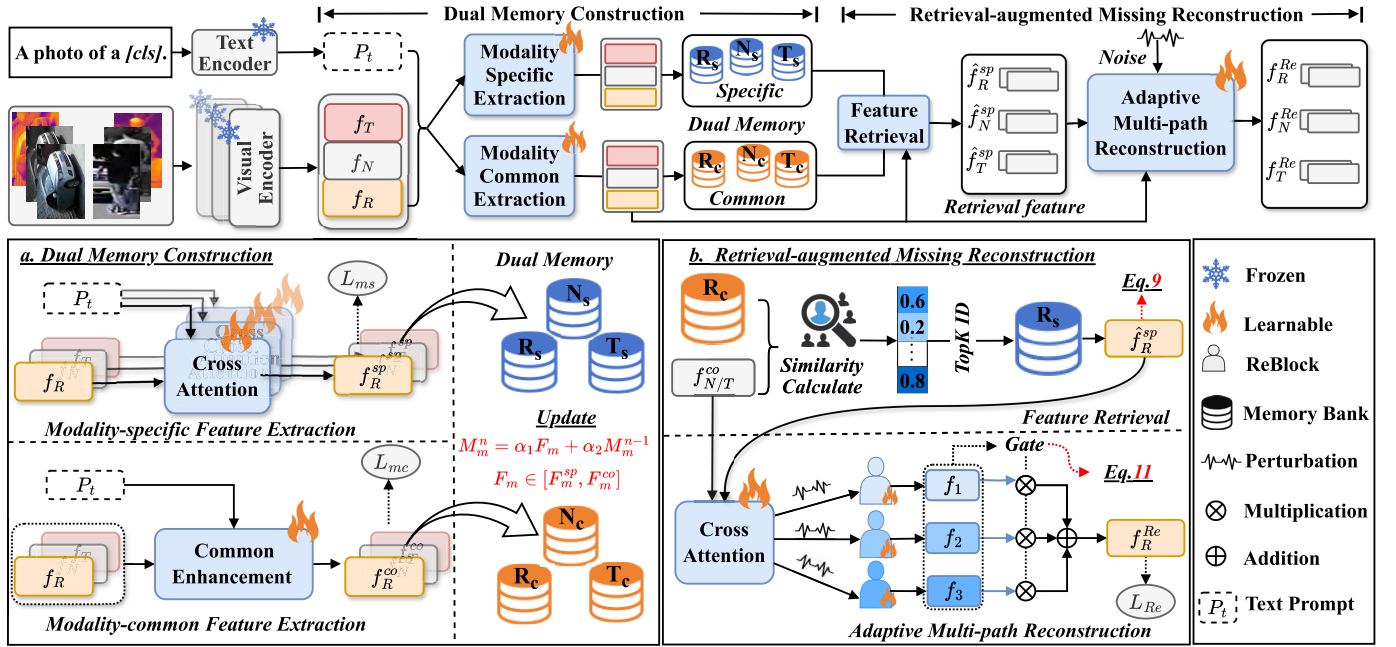


Fig. 2. Overview of the proposed REMIND framework. Image features are extracted using three independent visual encoders corresponding to RGB, NIR, and TIR spectra. As illustrated in Fig. 2 (a), the proposed Dual Memory Construction (DMC) disentangles modality-specific and modality-common features, which are stored separately in dedicated memory banks to serve as semantic priors. Taking the missing RGB modality for example, Fig. 2 (b) demonstrates the proposed Retrieval-Augmented Missing Reconstruction (RMR).

where $I(f_m^{sp}; F_{\text{fuse}})$ characterizes the dependency between the modality-specific feature and the fused feature, and $I(f_{m_1}^{co}, f_{m_2}^{co})$ characterizes the dependency between modality-common features from different modalities. These two dependencies provide useful insights for designing our regularization terms. Accordingly, we introduce one regularizer to suppress redundant coupling between modality-specific and fused features, and another to encourage semantic consistency among modality-common features.

1) **Modality-Specific Feature Extraction:** To enhance the discriminative capability of single-modal representations, we introduce fixed textual prompts to interact with image features, thereby guiding the network to focus on foreground information and refining modality-specific feature extraction:

$$f_m^{sp} = CA(P_t, f_m), \quad (4)$$

where $CA(\cdot)$ is Cross-Attention, and P_t is the textual prototype. To encourage modality-specific learning, we propose a modality-specific loss. The key idea is to leverage the variance of each modality as an adaptive weight in feature fusion. From an information-theoretic perspective, the entropy of a distribution increases with its variance, and this relation holds strictly for Gaussian distributions.

$$H(X) = \frac{1}{2} \log(2\pi e \sigma^2), \quad (5)$$

which is monotonic in the variance σ^2 . Therefore, variance can be regarded as a practical proxy for information richness. In addition, similar variance-based weighting schemes have been explored in multi-modal learning and uncertainty modeling [45], [46], [47].

Based on this principle, we normalize the variance of each modality as:

$$V_m = \frac{\sigma_m^2}{\sum_m \sigma_m^2 + 10^{-9}}, \quad m \in [R, N, T], \quad (6)$$

where σ_m^2 denotes the sample-wise variance of the feature vector of modality m , computed over the feature dimension for each sample in the current iteration, and serves as an adaptive weight to control each modality's contribution during feature fusion. The final adaptive fusion feature is computed as:

$$F_{\text{fuse}} = \sum_{m \in [R, N, T]} V_m f_m. \quad (7)$$

Inspired by information-theoretic principles, we aim to reduce the redundancy between the modality-specific feature f_m^{sp} and the fused feature F_{fuse} . Under the Gaussian assumption, the mutual information between correlated variables can be written as

$$I(f_m^{sp}; F_{\text{fuse}}) = -\frac{1}{2} \log(1 - \rho^2), \quad (8)$$

where ρ denotes their correlation coefficient. By applying a second-order Taylor expansion at $\rho = 0$, we obtain

$$\begin{aligned} I(f_m^{sp}; F_{\text{fuse}}) &= -\frac{1}{2} \log(1 - \rho^2) \\ &= 0 + 0 + \frac{1}{2} \rho^2 + o(\rho^2) \\ &\approx \frac{1}{2} \rho^2. \end{aligned} \quad (9)$$

This approximation suggests that reducing the correlation between f_m^{sp} and F_{fuse} can help suppress redundant common information. Motivated by this observation, we adopt an

orthogonality-inspired regularization on normalized features. Specifically, we apply L2 normalization:

$$\tilde{f}_m^{sp,i} = \frac{f_m^{sp,i}}{\|f_m^{sp,i}\|_2 + \epsilon}, \quad \tilde{F}_{\text{fuse}}^i = \frac{F_{\text{fuse}}^i}{\|F_{\text{fuse}}^i\|_2 + \epsilon}, \quad (10)$$

where ϵ is a small constant for numerical stability. We then define the modality-specific loss as:

$$\mathcal{L}_{ms} = \frac{1}{N} \sum_{i=1}^N \sum_{m \in M} V_m^i \left((\tilde{f}_m^{sp,i})^\top \tilde{F}_{\text{fuse}}^i \right)^2, \quad (11)$$

where N denotes the number of samples, M is the set of modalities, and V_m^i is the adaptive fusion weight of modality m for the i -th sample. By penalizing the squared inner product between L2-normalized modality-specific and fused features, this loss discourages directional alignment and promotes sample-level disentanglement between modality-specific and common information.

2) *Modality-Common Feature Extraction*: To extract robust and discriminative modality-common representations for reconstruction, and to facilitate the construction of a modality-common memory bank that alleviates the semantic heterogeneity in direct cross-modal transformations, we propose a Common Enhancement Module (CoEn) for unified representation learning. Unlike prior approaches that rely on simple concatenation or linear transformations, our design explicitly integrates semantic priors and structured modeling, enabling the extraction of common cues while mitigating modality discrepancies. Concretely, the module leverages fixed textual prompts P_t as semantic guidance to align heterogeneous features into a unified space. The enhanced features from different modalities are then concatenated and fed into a shared enhancement block to capture inter-modality complementarity. The process can be formulated as:

$$F^{\text{co}} = \text{CoEn}(\text{Cat}([f_R, f_N, f_T]), P_t), \quad \text{CoEn}(x_1, x_2) = \text{CA}(FFN(S(LN(x_1))), x_2), \quad (12)$$

where $F^{\text{co}} = \{f_R^{\text{co}}, f_N^{\text{co}}, f_T^{\text{co}}\}$, $\text{Cat}(\cdot)$ denotes concatenation along the token dimension ($\text{dim}=1$). Specifically, each modality-common feature is first represented as $f_R^{\text{co}}, f_N^{\text{co}}, f_T^{\text{co}} \in \mathbb{R}^{B \times 1 \times D}$, and then concatenated to obtain $F^{\text{co}} \in \mathbb{R}^{B \times 3 \times D}$, where B denotes the batch size and D denotes the feature dimension. S , LN , and FFN refer to self-attention, layer normalization, and feed-forward network, respectively, while CA represents cross-attention guided by the fixed textual prompt. In this way, modality-common features are aligned in a shared semantic space while being regularized by semantic prompts, resulting in robust and modality-agnostic representations.

To learn robust modality-common representations, we encourage semantic consistency between common features from different modalities. From an information-theoretic perspective, Eq. (9) suggests that stronger dependency between modality-common features is associated with more consistent shared semantics. Motivated by this observation, and further considering that modality-common representations are expected to be positively aligned in the feature space, we adopt cosine similarity as an alignment measure.

$$\cos(f_{m_1}^{\text{co},i}, f_{m_2}^{\text{co},i}) = \frac{(f_{m_1}^{\text{co},i})^\top (f_{m_2}^{\text{co},i})}{\|f_{m_1}^{\text{co},i}\|_2 \|f_{m_2}^{\text{co},i}\|_2}. \quad (13)$$

By maximizing the cosine similarity between modality-common features, the model is encouraged to align their common semantic directions across modalities. To additionally preserve feature-scale consistency and avoid degenerate over-alignment, we further introduce a mean squared error term to constrain their Euclidean distance. This joint constraint prevents excessive contraction of inter-modality differences while retaining discriminative information. The modality-common loss is then defined as

$$\mathcal{L}_{mc} = \frac{1}{N} \sum_{i=1}^N \left[(1 - \cos(f_{m_1}^{\text{co},i}, f_{m_2}^{\text{co},i})) + \eta \|f_{m_1}^{\text{co},i} - f_{m_2}^{\text{co},i}\|_2^2 \right], \quad (14)$$

where $m_1, m_2 \in [R, N, T]$, and η is a balancing coefficient that controls the trade-off between directional alignment and distance consistency, preventing the loss of discriminative cues caused by over-alignment.

3) *Memory Construction*: For the extracted modality-specific and modality-common features, we construct dedicated memory banks to store semantic priors. Specifically, taking the modality-specific branch as an example, we build three independent modality-specific memory banks, each with a size of $[N, D]$, where N denotes the total number of identities in the training set and D is the feature dimension. We perform a warm-up during the first epoch, and memory updates begin from the second epoch onward. After a warm-up epoch, memories update as:

$$M_{m,n}^{\text{sp/co}} \leftarrow \alpha F_m^{\text{sp/co}} + (1 - \alpha) M_{m,n-1}^{\text{sp/co}}, \quad (15)$$

where m denotes the modality types, n denotes the current training epoch, $\alpha \in [0, 1]$ is a momentum-like update coefficient.

C. Retrieval-Augmented Missing Reconstruction

1) *Feature Retrieval*: To reconstruct robust features for the missing modality, we propose a Retrieval-augmented Missing Reconstruction (RMR) mechanism. The core idea of RMR is to leverage the modality-common semantic space as a bridge for cross-modal retrieval, and then retrieve semantically similar information from the modality-specific memory bank of the missing modality. Specifically, when the target modality is absent, we first use the common features of the available modality to query the common memory bank of the missing modality. This retrieval process allows us to identify identity prototypes that are most semantically aligned with the query, thereby providing reliable priors for reconstruction.

Taking the case where the RGB modality is missing and the NIR modality is available, the cross-modal identity retrieval can be formulated as:

$$\mathcal{ID}_K = \text{TopK}(\cos(f_N^{\text{co}}, M_R^{\text{co}})), \quad \hat{f}_R^{\text{sp}} = \frac{1}{K} \sum_{i \in \mathcal{ID}_K} M_R^{\text{sp}}[i], \quad (16)$$

where $\mathcal{ID}_K$ denotes the index set of the top- K retrieved identities from the common memory M_R^{co} of the missing modality. The retrieval is performed by computing cosine similarity between the available modality's common feature f_N^{co} and

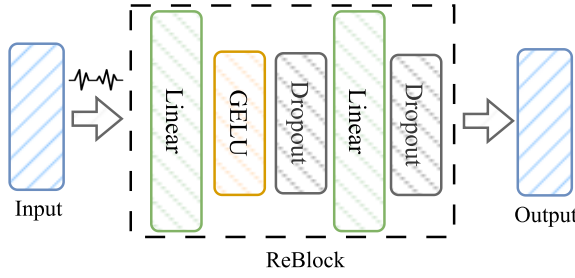


Fig. 3. One branch of multi-path reconstruction from available input to reconstructed feature.

all entries in M_R^{co} . The operator $\text{TopK}(\cdot)$ returns the indices corresponding to the top- K similarities. These indices are then used to access the corresponding modality-specific features $M_R^{sp}[i]$ stored in the missing modality's specific memory bank. Finally, the selected modality-specific features are averaged to obtain the reconstructed representation $\hat{f}_R^{sp}$.

2) *Adaptive Multi-Path Reconstruction*: Subsequently, the retrieved features from the missing modality and the common features from the available modality are fed into the multi-path reconstruction (MPR) module. Unlike conventional single-path reconstruction approaches, our multi-path design enables the network to capture diverse and complex semantic patterns from multiple reconstruction trajectories, thereby enhancing the expressiveness and robustness of the reconstructed representations. To further improve robustness under uncertain or noisy modality conditions, noise perturbation is injected into the input of each ReBlock. This strategy explicitly simulates modality uncertainty, encouraging the model to learn stable and invariant features that remain discriminative even when input modalities are incomplete or corrupted. The outputs from all reconstruction paths are then adaptively fused via a threshold-aware weighting mechanism to produce the final reconstructed feature. For instance, when the RGB modality is missing and NIR is available, the RMR module generates a set of candidate reconstructions:

$$F_R^{\text{Re}} = \{\text{ReBlock}_i(\psi(\mathbf{M}_{\text{CA}}(f_N^{co}, \hat{f}_R^{sp})))\}_{i=1}^K, \quad (17)$$

where $\mathbf{M}_{\text{CA}}$ denotes multi-head cross-attention [48], ψ represents noise perturbation, K indicates the number of ReBlocks. As shown in Fig. 3, each $\text{ReBlock}_i(\cdot)$ consists of two linear layers, a GELU activation, and dropout. The reconstructed features $F_R^{\text{Re}} = \{f_R^{\text{Re},1}, \dots, f_R^{\text{Re},K}\}$ serve as a rich set of reconstruction candidates. By combining multi-path reconstruction and noise-aware perturbation, our approach achieves a balance between discriminability and robustness, substantially improving performance under missing-modality scenarios.

To obtain more robust features, we compute adaptive weights over multiple reconstruction candidates:

$$G_m = \text{Linear}(\text{Soft}(F_m^{\text{Re}})),$$

$$f_m^{\text{Re}} = \sum_{i=1}^K G_m^i F_m^{\text{Re},i}, \quad (18)$$

where $F_m^{\text{Re}} = [f_m^{\text{Re},1}, f_m^{\text{Re},2}, \dots, f_m^{\text{Re},K}]$, $m \in [R, N, T]$ is the set of reconstructed features from K ReBlock branches, Soft represents the softmax function.

Finally, we apply a mean squared error (MSE) loss between the reconstructed feature and the ground-truth feature of the missing modality, enforcing proximity in the feature space to ensure the fidelity of the reconstruction.

$$\mathcal{L}_{\text{Re}} = \|f_m^{\text{Re}} - f_m\|_2^2. \quad (19)$$

By minimizing the reconstruction loss $\mathcal{L}_{\text{Re}}$, we constrain the distance between the reconstructed feature and the ground-truth feature, thereby enhancing the robustness of the reconstruction.

D. Loss Function

As illustrated in Fig. 2, the training objective comprises a combination of cross-entropy ($\mathcal{L}_{ce}$) loss and triplet ($\mathcal{L}_{tri}$) loss to supervise identity learning:

$$\mathcal{L}_g = \mathcal{L}_{ce} + \mathcal{L}_{tri}. \quad (20)$$

In addition, the proposed modality-specific loss, modality-common loss, and reconstruction loss are jointly optimized to guide the overall training of the REMIND framework:

$$\mathcal{L}_{\text{task}} = \mathcal{L}_g + \lambda \mathcal{L}_{ms} + \mathcal{L}_{mc} + \mathcal{L}_{\text{Re}}. \quad (21)$$

Here, λ is a tunable hyperparameter controlling the weight of the modality-specific loss. While $\mathcal{L}_{ms}$ encourages orthogonality to reduce redundancy, an overly large λ may suppress discriminative cues. Thus, λ is set to balance modality-specific preservation and identity-discriminative capability.

The pseudo-code procedure of the proposed REMIND strategy is as follows:

Algorithm 1 The Proposed Algorithm for REMIND

Require: Input feature f ; Fixed textual prompt p

Ensure: Reconstructed feature f_m^{Re}

```

1: for  $i = 1$  to  $epochs$  do
2:    $f_m^{sp}$  extraction via Eq. (4) and (11)
3:    $f_m^{co}$  extraction via Eq. (12) and (14)
4:   if  $i > 1$  then
5:     Update dual memories  $M_{m,n}^{sp/co}$  via Eq. (15)
6:    $f_m^{sp}$  retrieval via Eq. (16)
7:    $f_m^{\text{Re}}$  reconstruction via Eq. (17), (18) and (19)
8:   end if
9:    $f$  update via Eq. (20)
10:  Update the encoder weight by optimizer
11: end for

```

IV. EXPERIMENTS

A. Dataset and Evaluation Protocols

1) *Datasets*: We evaluate our proposed framework on four benchmark datasets for multi-modal object Re-ID: RGBNT100 [1] consists of 17,250 images from 100 vehicle identities, with 50 identities for training and the remaining 50 identities for testing. MSVR310 [3] contains 2,087 images from 310 vehicle identities, where 155 identities are used

for training and the other 155 identities are used for testing. WMVEID863 [4] is a multi-modal vehicle Re-ID dataset with 4,709 images from 863 vehicle identities captured across 8 cameras, including 575 identities for training and 288 identities for testing. It is also the first multi-spectral vehicle Re-ID dataset designed for intense illumination scenarios. RGBNT201 [2] is a multi-modal person Re-ID dataset with 4,787 images from 201 person identities, including 171 identities for training and the remaining 30 identities for testing. All four datasets are organized as RGB–NIR–TIR image triplets.

2) *Evaluation Metrics*: We assess our Re-ID model using two widely adopted metrics: Cumulative Matching Characteristic (CMC) and mean Average Precision (mAP). The CMC metric evaluates retrieval performance by reporting the likelihood of correctly matching an identity within the top- k ranked results, specifically focusing on R-1, R-5, and R-10 accuracies. The mAP metric reflects both precision and recall by averaging the precision over all recall levels for each query.

B. Missing Setting and Implementation Details

1) *Modality-Missing Scenario Settings*: (1) Fixed Missing: For input pairs of test samples, one or two modalities are fixed missing. (2) Random Missing: For the input test sample pairs, the samples are randomly missing at a certain ratio based on the random missing setting in CCNet [3], which indicates the probability of missing samples in random partial modality. It is important to note that all missing strategies are applied during testing, while the model is trained on complete modality data. The tests are conducted under different missing settings to evaluate performance under various scenarios.

2) *Implementation Details*: We implement our method in PyTorch and train on a single NVIDIA GeForce RTX 4090 GPU. Under the complete-modality setting, we employ CLIP visual encoders pretrained on ImageNet [51] as the backbone architecture. Notably, since multi-modal person datasets exhibit better performance with common feature extractors, we adopt parameter sharing for the visual encoders on the RGBNT201 [2] dataset. For vehicle Re-ID tasks, input images are resized to 128×256 , while for person Re-ID tasks, images are resized to 256×128 . Standard data augmentation techniques, including random cropping and horizontal flipping, are applied during training. The Adam optimizer is used to optimize the network, with an initial learning rate of 3.5×10^{-4} . For the WMVEID863 dataset specifically, we set a smaller initial learning rate of 5×10^{-6} . Specifically, the models are trained for 20 epochs on RGBNT100, 70 epochs on MSVR310, and 120 epochs on both WMVEID863 and RGBNT201. We follow the common practice in multi-modal Re-ID [12], [13] and periodically evaluate candidate checkpoints during training. For each dataset, the best-performing checkpoint is selected only within its predefined training schedule and recorded as the final model for reporting. For the setting of λ , we empirically adopt different values across datasets due to variations in data distribution. Specifically, we set $\lambda = 0.5$ for RGBNT100, and $\lambda = 0.1$ for MSVR310, WMVEID863, and RGBNT201. Unless otherwise stated, “M (·)” stands for missing modality in the following tables, and the best and second-best results are highlighted in **bold** and underlined, respectively.

C. Comparison With State-of-the-Art Methods

1) *Evaluation on Modality-Missing Scenarios: Fixed Missing*: As shown in Table I, under the fixed modality-missing conditions of four benchmark datasets, REMIND outperforms the existing state-of-the-art methods, achieving the best results across all evaluation metrics. Notably, REMIND demonstrates greater robustness on the WMVEID863 [4], which has strong lighting conditions, highlighting its excellent performance in scenarios similar to partial modality loss, such as in strong lighting conditions. Furthermore, REMIND also shows significantly better performance under extreme missing conditions (where two modalities are missing) across multiple datasets. We observe that the performance varies under different missing configurations, which is due to the significant modality imbalance between the different modalities. This result validates the importance of the orthogonality constraint in preserving modality-specific discriminative cues.

Random Missing: To further evaluate its generalization, we also tested the model’s performance under randomly missing modalities at varying proportions. For each missing ratio, we conduct 10 independent runs on the same NVIDIA GeForce RTX 4090 GPU and report the average results as the final performance. As shown in Table II, under different missing proportions, particularly in high-missing-rate conditions (where the missing rate is greater than 0.5), REMIND significantly outperforms the state-of-the-art model DeMo [13] and PromptMA [14] on the multi-modal vehicle dataset, and also shows a considerable advantage on the multi-modal person dataset. These results confirm that REMIND demonstrates robustness and effectiveness in addressing the modality incompleteness issue in multi-modal object Re-ID.

2) *Evaluation on Complete Modality Scenarios*: As shown in Table III, although REMIND was originally designed to address the modality-missing problem in multi-modal Re-ID, maintaining high recognition performance when some modalities are unavailable, experimental results demonstrate that it also achieves highly competitive performance under the complete-modality setting. Under the complete-modality setting, the RMR branch is not used during inference, and the final retrieval feature is obtained by concatenating the modality-specific and modality-common features extracted by DMC. Across multiple benchmark datasets, REMIND consistently ranks at the top, and it exhibits comparable or even superior results on the multi-modal vehicle dataset MSVR310 [3]. This indicates that the proposed design is not limited to missing-modality scenarios but also possesses strong generalization ability and robustness. This advantage primarily stems from two key components within the model. First, the DMC module enhances the learning of robust fine-grained modality-specific representations and modality-common representations, thereby improving feature discriminability and semantic consistency. Second, although the RMR branch is not executed during complete-modality inference, it still contributes during training as a reconstruction regularization mechanism, which indirectly alleviates cross-modal heterogeneity by constraining the alignment of reconstructed features. Together, these mechanisms allow REMIND to maintain robustness under complete-modality conditions while further

TABLE I
EXPERIMENTAL RESULTS UNDER FIXED MODALITY-MISSING CONDITIONS

RGBNT100														
Methods	M (RGB)		M (NIR)		M (TIR)		M (RGB+NIR)		M (RGB+TIR)		M (NIR+TIR)		Average	
	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1
CCNet [3]	66.8	90.2	73.2	92.2	60.0	82.9	44.4	75.0	42.4	63.8	49.5	69.8	56.1	79.0
TOP-ReID [12]	70.6	90.6	77.9	94.5	64.0	81.5	42.5	69.3	45.9	65.4	55.4	77.8	59.4	80.0
MDReID [49]	79.0	91.0	83.4	94.5	67.9	80.6	46.1	69.7	56.4	71.1	61.8	79.9	65.8	81.1
PromptMA [14]	80.3	95.8	83.5	96.8	69.3	85.8	48.0	74.3	56.2	75.2	64.2	82.0	66.9	85.0
DeMo [13]	81.0	94.5	84.1	96.5	71.1	87.6	50.2	73.7	59.6	78.1	66.3	82.8	68.7	85.5
REMIND (Ours)	82.5	96.6	87.4	97.9	74.9	90.1	53.8	79.9	61.7	79.8	66.9	83.3	71.2	87.9
MSVR310														
CCNet [3]	25.5	43.7	27.2	42.3	26.4	41.6	10.8	30.1	20.6	34.2	24.1	37.2	22.4	38.2
TOP-ReID [12]	23.5	41.1	28.6	41.8	31.1	45.9	10.8	23.5	22.3	40.6	26.5	36.5	23.8	38.2
DeMo [13]	36.9	55.3	43.1	56.5	46.1	60.9	10.5	24.2	34.1	53.5	40.8	53.6	35.3	50.7
MDReID [49]	40.2	55.7	46.2	63.8	45.6	64.5	14.4	31.6	33.4	52.2	42.9	58.0	37.1	54.3
REMIND (Ours)	41.1	57.2	49.2	66.5	48.4	67.0	20.8	34.5	34.9	52.5	45.5	60.7	40.0	56.4
WMVEID863														
CCNet [3]	58.7	64.8	59.8	64.9	53.5	58.7	51.9	56.8	43.0	44.1	46.7	48.6	52.3	56.3
TOP-ReID [12]	63.2	69.9	66.8	73.7	55.3	59.8	60.3	65.7	46.8	49.5	53.0	56.4	57.6	62.5
DeMo [13]	66.5	73.3	65.9	73.3	55.5	59.8	58.4	63.9	49.7	52.1	50.4	52.8	57.7	62.5
MDReID [49]	67.0	75.3	70.1	78.3	59.4	65.3	59.4	67.6	51.1	54.6	56.7	61.7	60.6	67.1
REMIND (Ours)	70.1	77.8	71.9	79.0	61.2	68.5	61.9	66.4	54.6	59.1	58.6	64.2	63.1	69.2
RGBNT201														
PFNet [2]	-	-	31.9	29.8	25.5	25.8	-	-	-	-	26.4	23.4	-	-
TOP-ReID [12]	54.4	57.5	64.3	67.6	51.9	54.5	35.3	35.4	26.2	26.0	34.1	31.7	44.4	45.5
DeMo [13]	63.3	65.3	72.6	75.7	56.2	54.1	45.6	46.5	26.3	24.9	40.3	38.5	50.7	50.8
MDReID [49]	67.0	67.9	75.8	80.9	61.7	59.8	48.6	51.1	31.4	29.2	42.1	40.0	54.4	54.8
Miss-ReID [50]	66.6	68.2	72.4	75.5	63.2	63.8	47.2	49.5	34.5	33.3	43.9	44.3	54.6	55.7
PromptMA [14]	67.4	68.4	72.5	75.7	58.9	57.3	51.5	53.0	30.4	33.3	43.9	42.2	54.1	55.0
REMIND (Ours)	71.6	71.9	76.7	81.1	63.9	63.5	53.5	55.4	32.3	35.6	47.1	47.5	57.5	59.2

TABLE II
EXPERIMENTAL RESULTS UNDER PERCENTAGE-BASED RANDOM MODALITY-MISSING CONDITIONS

RGBNT100													
Methods	0.1		0.3		0.5		0.7		0.9		Average		
	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	
PromptMA [14]	69.7	93.6	46.9	92.1	28.4	<u>90.1</u>	19.8	88.5	15.2	85.7	36.0	<u>90.0</u>	
DeMo [13]	<u>77.1</u>	<u>94.8</u>	41.2	91.3	<u>31.9</u>	88.7	23.0	87.5	<u>17.6</u>	<u>86.1</u>	<u>38.2</u>	89.7	
REMIND (Ours)	81.6	96.9	67.5	96.7	55.5	96.0	45.5	95.3	37.9	94.8	57.6	95.9	
MSVR310													
PromptMA [14]	35.6	54.2	26.1	48.3	14.7	36.9	10.8	31.1	9.3	29.0	19.3	39.9	
DeMo [13]	<u>42.5</u>	<u>60.6</u>	<u>32.4</u>	<u>50.1</u>	<u>23.9</u>	<u>44.5</u>	<u>19.4</u>	<u>38.2</u>	<u>18.2</u>	<u>39.7</u>	<u>27.3</u>	<u>46.6</u>	
REMIND (Ours)	50.7	66.7	44.0	60.3	38.1	55.2	32.6	49.5	28.3	45.2	38.7	55.4	
WMVEID863													
PromptMA [14]	59.1	69.6	33.2	43.7	21.8	36.0	15.3	24.9	11.6	20.6	28.2	39.0	
DeMo [13]	<u>65.4</u>	75.8	<u>34.0</u>	<u>45.6</u>	<u>25.4</u>	<u>39.5</u>	<u>18.6</u>	<u>29.1</u>	<u>13.0</u>	<u>24.7</u>	<u>31.3</u>	<u>42.9</u>	
REMIND (Ours)	67.6	<u>74.0</u>	61.8	66.8	56.1	61.3	48.2	52.8	39.1	47.3	54.6	60.4	
RGBNT201													
PromptMA [14]	<u>63.2</u>	<u>71.1</u>	38.9	<u>62.4</u>	25.7	57.6	16.3	55.0	10.8	41.2	31.0	<u>57.5</u>	
DeMo [13]	60.7	69.3	<u>40.2</u>	56.6	<u>32.9</u>	60.4	<u>23.8</u>	51.7	<u>15.5</u>	46.0	<u>34.6</u>	56.8	
REMIND (Ours)	71.3	76.6	53.6	68.0	39.0	<u>59.9</u>	26.9	<u>52.3</u>	18.6	<u>45.2</u>	41.9	60.4	

leveraging its strengths in feature alignment and refinement, state-of-the-art methods specifically optimized for complete-modality scenarios. resulting in overall performance that surpasses or matches modality scenarios.

TABLE III
COMPARISON OF DIFFERENT METHODS UNDER THE
COMPLETE MODALITY SETTING

Methods	RGBNT100		RGBNT201		MSVR310		WMVEID863	
	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1
ICPL [37]	87.0	98.6	75.1	77.4	56.9	77.7	-	-
PromptMA [14]	85.3	97.4	78.4	80.9	<u>55.2</u>	64.5	67.1	73.5
DeMo [13]	86.2	97.6	79.0	82.3	49.2	59.8	68.8	<u>77.2</u>
MambaPro [34]	83.9	94.7	78.9	83.4	47.0	56.5	69.5	<u>76.9</u>
IDEA [9]	<u>87.2</u>	96.5	<u>80.2</u>	82.1	47.0	62.4	-	-
REMIND	88.0	98.7	81.0	84.2	51.1	<u>70.6</u>	70.4	78.1

D. Ablation Study

To evaluate the robustness of the proposed REMIND framework under missing-modality scenarios, we conduct ablation studies on multi-modal person and vehicle Re-ID datasets using fixed missing-modality settings (i.e., only available modalities are used during testing). As shown in Table IV, introducing the DMC module enhances single-modality feature extraction and significantly improves robustness over the baseline. Combined with the RMR module, which reconstructs missing features from available inputs and memory priors, REMIND achieves consistent gains across various missing cases. Notably, under extreme conditions where only one modality is available, it still delivers strong performance. In addition, under the complete-modality setting, the two proposed modules also bring performance gains to different extents. This can be attributed to the robust fine-grained modality-specific representations and modality-common representations learned by the DMC module, as well as the fact that RMR indirectly alleviates cross-modal heterogeneity by constraining the alignment of reconstructed features during training.

E. Discussions of DMC

1) *Effect of Feature Extraction Loss*: As shown in Table V, we conduct ablation studies to evaluate the impact of each component in modality-specific loss $\mathcal{L}_{ms}$ and modality-common loss $\mathcal{L}_{mc}$. The table reports the average results under the fixed-missing setting across different missing-modality combinations. “Variance” and “Average” denote variance- and mean-based fusion strategies. Results show variance fusion notably improves $\mathcal{L}_{ms}$, highlighting its effectiveness in capturing modality-specific cues. For $\mathcal{L}_{mc}$, removing directional (cosine) or distance (MSE) constraint degrades performance, indicating both are important for learning robust common features. Overall, $\mathcal{L}_{ms}$ and $\mathcal{L}_{mc}$ are essential to DMC, enabling discriminative, reliable feature extraction under modality-missing conditions.

2) *Effect of Memory Bank Setting*: To further analyze the role of memory bank settings in REMIND, we conduct ablation studies on perturbed memory bank writing features, retrieval strategies, and effective memory size on RGBNT201 [2], as shown in Table VI. Table VI (a) shows that perturbing the features written into the memory bank degrades performance compared with the full REMIND setting in Table VI (d), indicating that retrieval-guided reconstruction

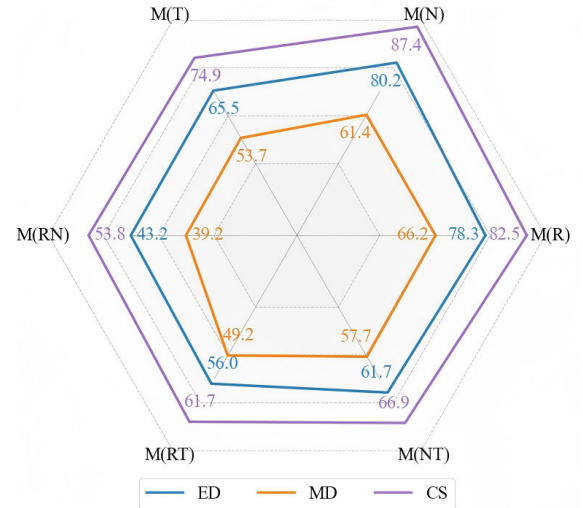


Fig. 4. Reconstruction results of different retrieval methods under fixed missing patterns on the RGBNT100. ED denotes Euclidean Distance, MD denotes Mahalanobis Distance, and CS denotes Cosine Similarity.

relies on stable stored features. Table VI (b) shows that similarity-based retrieval outperforms both random retrieval and the variant without retrieval, suggesting that the gain of REMIND comes from semantically relevant retrieved features rather than merely introducing a memory mechanism. Table VI (c) shows that performance improves as N increases from 20 to 100, while the overall variation remains limited, indicating low sensitivity to memory size. Overall, these results suggest that the proposed dual-memory mechanism serves as a structured semantic prior rather than memorization of the training set.

3) *Evaluation of Text Prompt Enhancement*: As shown in Table VII, we conduct ablation experiments under different fixed missing-modality settings on two datasets to evaluate the effect of text prompt enhancement. Overall, introducing text prompts improves the average performance over the variant without text on both datasets, showing that textual guidance provides useful complementary semantic cues for modality-missing multi-modal Re-ID. Among them, fixed prompts achieve the best average performance, while learnable prompts also bring gains but remain inferior overall. This suggests that a simple fixed prompt template provides more stable and effective semantic guidance under the current setting. Although learnable prompts are more flexible, they may introduce additional optimization difficulty on limited training data. Nevertheless, learnable prompts and more descriptive textual guidance remain promising directions for future work.

F. Discussions of RMR

1) *Evaluation of Top-K Retrieval*: As shown in Fig. 4, under fixed modality-missing conditions, we compare different retrieval strategies’ impact on reconstructed feature quality. Specifically, we evaluate Euclidean Distance (ED), Mahalanobis Distance (MD), and Cosine Similarity (CS) as retrieval metrics. The results show that using Cosine Similarity to retrieve Top-K nearest samples leads to more accurate and discriminative reconstructions. This suggests angular similarity

TABLE IV
ABLATION RESULTS UNDER DIFFERENT MODALITY-MISSING COMBINATIONS ON RGBNT100 AND RGBNT201 (IN %)

RGBNT100																
Methods	Complete		M (RGB)		M (NIR)		M (TIR)		M (RGB+NIR)		M (RGB+TIR)		M (NIR+TIR)		Average	
	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1
Baseline	85.3	96.5	78.4	94.5	84.7	97.6	71.4	85.6	48.7	70.5	58.7	74.3	60.6	76.7	67.6	83.2
+DMC	87.2	97.6	80.0	95.3	84.8	97.4	74.3	87.9	48.9	74.3	59.3	77.3	66.7	80.1	69.0	85.4
+DMC+RMR	88.0	98.7	82.5	96.6	87.4	97.9	74.9	90.1	53.8	79.9	61.7	79.8	66.9	83.3	71.2	87.9

RGBNT201																
Methods	Complete		M (RGB)		M (NIR)		M (TIR)		M (RGB+NIR)		M (RGB+TIR)		M (NIR+TIR)		Average	
	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1	mAP	R-1
Baseline	78.0	80.0	66.5	66.7	73.8	77.4	54.2	51.4	50.6	52.4	29.4	25.8	37.1	33.9	51.9	51.3
+DMC	79.8	81.8	67.2	66.3	73.9	76.0	60.8	58.6	51.1	52.9	32.7	32.5	39.8	35.3	54.3	53.6
+DMC+RMR	81.0	84.2	71.6	71.9	76.7	81.1	63.9	63.5	53.5	55.4	32.3	35.6	47.1	47.5	57.5	59.2

TABLE V
COMPARISON OF DIFFERENT LOSS CONFIGURATIONS IN DMC ON THE RGBNT201 AND WMVEID863 DATASETS (IN %)

$\mathcal{L}_{ms}$		$\mathcal{L}_{mc}$		RGBNT201		WMVEID863	
Variance	Average	COS	MSE	mAP	R-1	mAP	R-1
✓	×	✓	×	51.9	53.9	62.4	67.3
✓	×	×	✓	52.6	52.4	62.0	67.6
×	✓	✓	✓	56.1	60.4	61.4	66.8
✓	×	✓	✓	57.5	59.2	63.1	69.2

TABLE VI
ABLATION STUDY OF DIFFERENT MEMORY BANK SETTINGS ON RGBNT201 (IN %)

Methods	Setting	mAP	R-1
(a) Noise Perturbation	Weak Noise	56.4	55.9
	Strong Noise	55.3	53.4
(b) Retrieval Strategy	Random Retrieval	53.9	55.2
	w/o Retrieval	55.4	57.6
(c) Effective Memory Size	$N = 20$	56.7	58.1
	$N = 60$	56.9	58.6
	$N = 100$	57.4	59.0
(d) Ours	REMIND	57.5	59.2

TABLE VII
ABLATION RESULTS OF TEXT PROMPTS ON WMVEID863 AND RGBNT201 DATASETS (IN %)

Setting	WMVEID863			RGBNT201		
	w/o text	Learn.	Fixed	w/o text	Learn.	Fixed
M (R)	68.7	68.5	70.1	67.5	69.5	71.6
M (N)	70.5	70.1	71.9	74.6	78.1	76.7
M (T)	62.6	61.6	61.2	61.7	61.8	63.9
M (RN)	55.3	59.6	61.9	49.7	53.2	53.5
M (RT)	53.5	54.5	54.6	26.2	28.1	32.3
M (NT)	59.0	57.9	58.6	36.2	45.2	47.1
Average	61.2	62.0	63.0	53.0	56.0	57.5

better captures semantic consistency between samples, making it more suitable for guiding feature recovery in heterogeneous modality spaces.

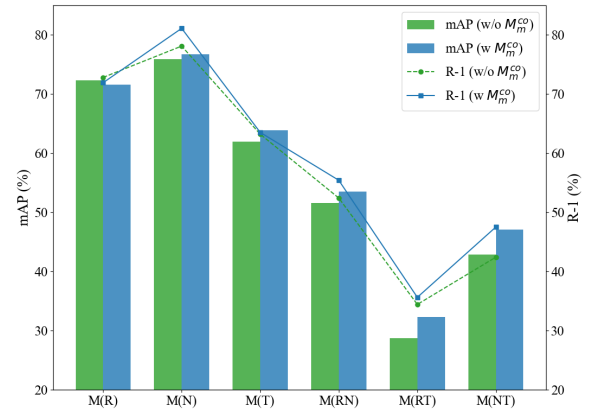


Fig. 5. Ablation Study of Cross-modal Retrieval with or without Common Memory Bank on RGBNT201.

2) *Evaluation of Cross-Modal Feature Retrieval:* Fig. 5 demonstrates the effectiveness of our retrieval strategy in mitigating cross-modal heterogeneity in multi-modal Re-ID. Compared to existing retrieval methods that directly query modality-specific memory banks, our approach first retrieves from a cross-modal common memory bank to obtain the corresponding ID labels, and then uses these IDs to extract modality-specific features from dedicated memory banks. This two-step retrieval process allows the model to better leverage semantically aligned, cross-modally consistent information before accessing modality-specific details. As a result, it significantly improves the quality and discriminative power of the reconstructed features. This strategy effectively alleviates heterogeneity among modalities and enhances the accuracy and robustness of feature reconstruction. These results validate the critical role of our retrieval method in advancing multi-modal representation learning.

3) *Effects of the Number of Top-K and ReBlocks:* To assess the effect of the number of ReBlocks on reconstruction, we conduct experiments on four datasets with the number of ReBlocks K ranging from 2 to 7, as shown in Fig. 6 (a). The results show that the performance generally improves as K increases and reaches the best performance around $K = 3$ or $K = 4$ on different datasets. This suggests that multi-path reconstruction helps capture more discriminative information

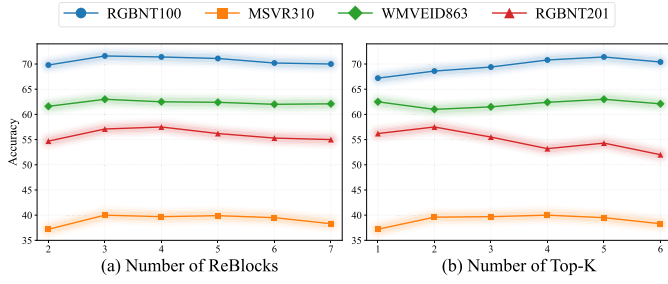


Fig. 6. Effect of ReBlock Quantity and Retrieved Feature Count on Reconstruction Results (Comparison of Average Values Under Fixed Modality Missing Conditions Across Different Datasets).

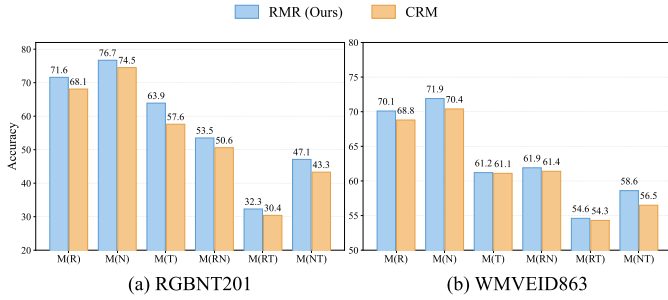


Fig. 7. The proposed RMR module and the existing missing value reconstruction method CRM [12] are compared on the WMVEID863 and RGBNT201 across all fixed missing modalities.

from diverse perspectives. In addition, the choice of Top-K retrieval samples also plays an important role in feature reconstruction. We conduct ablation experiments on the four datasets with Top-K values ranging from 1 to 6, as shown in Fig. 6 (b). The results show that, on the multi-modal vehicle datasets, the performance is generally better when Top-K is set to 4 or 5. In contrast, on the multi-modal person datasets, a smaller Top-K tends to achieve better performance, likely because these datasets exhibit stronger intra-class variation and more discriminative fine-grained attributes. These observations highlight the importance of selecting an appropriate Top-K strategy for different data characteristics.

4) *Effect of Different Reconstruction Methods*: To further validate the effectiveness of the proposed Retrieval-augmented Missing Reconstruction (RMR) module, we compare it with a mainstream reconstruction method. The CRM method [12] employs a single-path transformation approach, where available modality features are converted into missing modality features. As shown in Fig. 7, we conduct tests in various missing modality scenarios on both the multi-modal person dataset RGBNT201 [2] and the multi-modal vehicle dataset WMVEID863 [4]. The results demonstrate that the proposed RMR module consistently outperforms the single-path reconstruction method across different missing scenarios in various Re-ID tasks.

5) *Effect of Perturbation on Reconstruction*: To evaluate the effect of input perturbations on reconstruction, we train under three settings: no perturbation and two perturbation types. As shown in Fig. 8, applying noise to the input achieves the best results, indicating that proper noise-based perturbation enhances the robustness and quality of reconstructed features.

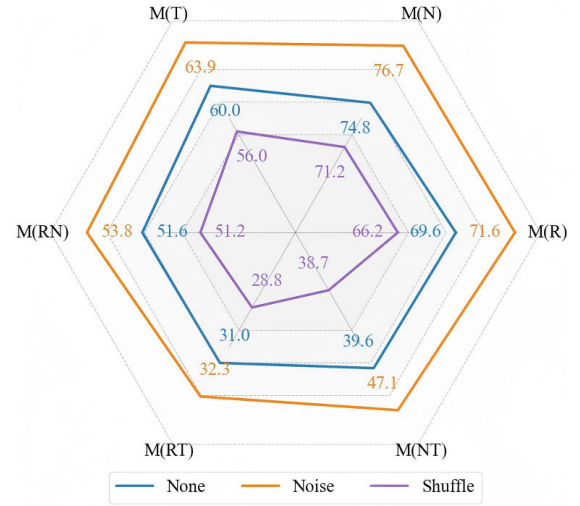


Fig. 8. Comparative analysis of reconstruction on the RGBNT201 dataset using input features without perturbation, with noise, and with shuffled perturbation.

TABLE VIII

COMPARISON WITH REID SOTA METHODS AND ABLATIONS OF RECONSTRUCTION METHODS, REBLOCKS (K), AND MEMORY BANK SIZE (N) ON RGBNT201

(a) ReID SOTA methods							
Methods				Params (M)	FLOPs (G)	RGBNT201 mAP	R-1
TOP-ReID [12]				324.5	35.5	44.4	45.5
DeMo [13]				98.8	35.1	50.7	50.8
MDReID [49]				57.2	22.3	54.4	54.8
Miss-ReID [50]				89.6	43.6	54.6	55.7
PromptMA [14]				107.9	36.2	54.1	55.0
REMIND (Ours)				118.6	37.1	57.5	59.2

(b) Reconstruction methods							
Methods	K	N	Latency (ms)	Params (M)	FLOPs (G)	RGBNT201 mAP	R-1
CRM	-	-	34.0	108.1	37.3	54.1	54.8
RMR	1	171	32.1	106.0	36.3	54.3	54.8
	2	171	32.4	110.2	36.6	54.7	55.6
	3	171	33.1	114.4	36.8	56.1	57.5
	4	20	32.0	118.6	37.0	56.7	58.1
	4	100	32.7	118.6	37.1	57.4	59.0
	4	171	33.5	118.6	37.1	57.5	59.2
	5	171	33.8	122.8	37.3	55.7	56.7

G. Efficiency and Hyperparameter Analysis

1) *Efficiency Analysis*: To evaluate the practical overhead of the proposed dual-memory retrieval and multi-path reconstruction, Table VIII reports the number of parameters, FLOPs, and inference latency on RGBNT201 [2]. Compared with existing state-of-the-art ReID methods, REMIND achieves the best performance while maintaining moderate computational cost, indicating limited inference overhead. We further compare the proposed RMR with the single-path CRM [12] and analyze the effects of the number of ReBlocks K and the effective memory size N . As shown in Table VIII (b), RMR consistently outperforms CRM [12]. Increasing K improves accuracy with only moderate increases in parameters and FLOPs, while the latency remains within a narrow range. When fixing $K = 4$ and varying N , the performance improves as N increases, whereas the latency rises slightly and the FLOPs remain nearly

TABLE IX
HYPERPARAMETER ANALYSIS ON RGBNT201 AND WMVEID863

Hyper-parameters	WMVEID863		RGBNT201	
λ ($\alpha = 0.2$)	mAP	R-1	mAP	R-1
0.1	63.1	69.2	57.5	59.2
0.3	62.8	71.1	55.9	57.3
0.5	62.2	70.8	55.4	56.8
0.7	61.4	69.7	52.3	54.1
0.9	61.7	70.1	53.7	54.9
1.0	61.1	69.5	51.4	52.3
α ($\lambda = 0.1$)	mAP	R-1	mAP	R-1
0.1	61.5	69.7	55.5	55.5
0.2	63.0	69.2	57.5	59.2
0.3	62.6	70.2	56.9	58.5
0.4	62.3	71.3	55.7	57.3
0.5	61.8	70.3	56.5	58.9

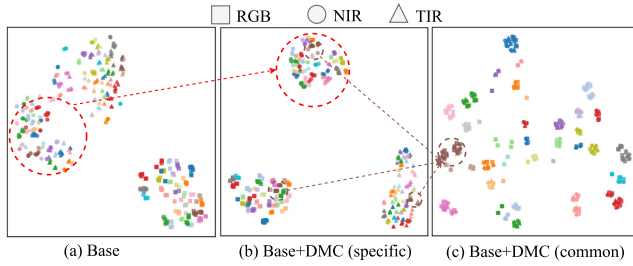


Fig. 9. Visualization of t-SNE on RGBNT100. (a) Baseline Feature Distributions. (b) Modality-Specific Feature Distributions. (c) Modality-Common Feature Distributions.

unchanged. These results indicate that REMIND benefits from richer memory candidates while keeping the overall inference cost well controlled.

2) *Hyperparameter Analysis*: To evaluate the sensitivity of key hyperparameters, we conduct ablation studies on the balancing coefficient λ in the modality-specific information extraction loss and the memory update coefficient α on WMVEID863 [4] and RGBNT201 [2]. We vary λ from 0.1 to 1.0 and compare the average performance under the fixed-missing setting on the two datasets. As shown in Table IX, the model achieves the best mAP on both datasets when $\lambda = 0.1$. Compared with WMVEID863 [4], RGBNT201 [2] shows higher sensitivity to λ . We further analyze the effect of α . As shown in Table IX, the model remains relatively stable over different α values, and $\alpha = 0.2$ achieves the best mAP on both datasets. Similar to λ , RGBNT201 [2] is also more sensitive to α than WMVEID863 [4].

H. Visualization Analysis

1) *Dual Feature Distributions*: Fig. 9 shows the t-SNE [52] distributions of features extracted by the baseline network, as well as the modality-specific and modality-common features extracted by the proposed DMC module. The visualization reveals that the modality-specific features obtained via DMC exhibit clear separation across different modalities, demonstrating strong modality disentanglement. In contrast, the modality-common features are distributed more compactly

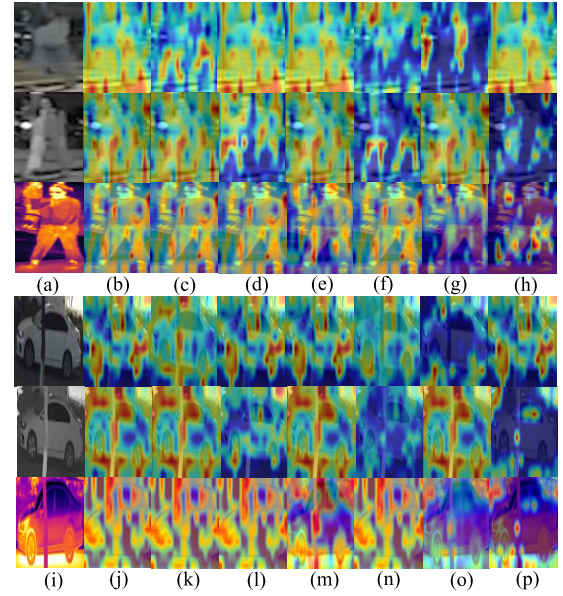


Fig. 10. The visualized results of the attention map. (a) Inputs, (b) complete, (c) M (RGB), (d) M (NIR), (e) M (TIR), (f) M (RGB+NIR), (g) M (RGB+TIR), (h) M (NIR+TIR). (i)–(p) show the corresponding results on the vehicle dataset under the same missing-modality settings.

across modalities, indicating that they effectively capture discriminative information that is consistent and shared across modalities.

2) *Attention Map Visualization*: As shown in Fig. 10, we visualize the Grad-CAM [53] of features extracted by REMIND on the RGBNT201 and RGBNT100 datasets. Fig. 10 (a) shows the input images, and Fig. 10 (b) presents the attention maps under the complete-modality setting. Although the attended regions vary across modalities, they consistently focus on discriminative parts of the person. Fig. 10 (c)–(e) show cases where the R, N, and T modalities are missing, respectively. In these cases, the model reconstructs the missing modality using the remaining two modalities and the modality-specific memory bank, while the resulting attention maps still highlight discriminative regions. Fig. 10 (f)–(h) illustrate more challenging cases with only one modality available, where the model still recovers part of the discriminative regions, demonstrating its robustness. Fig. 10 (i)–(p) show similar results on the vehicle dataset, where REMIND consistently reconstructs meaningful attention regions under different missing-modality settings. Even in extreme missing cases, the model still recovers partially informative regions. These visualizations further demonstrate the effectiveness and adaptability of REMIND for modality-missing object Re-ID.

3) *Discussion of Limitations*: As shown in Fig. 11, we visualize failure cases under extreme missing-modality scenarios on WMVEID863 [4] and RGBNT201 [2]. Specifically, (a)–(d) show reconstruction failures on strongly illuminated samples in WMVEID863 [4]. When the available modality is severely degraded by strong illumination, it provides limited discriminative information, which leads to less discriminative reconstructed missing-modality features. Similarly, (e)–(h) show comparable cases on RGBNT201 [2], where poor-quality

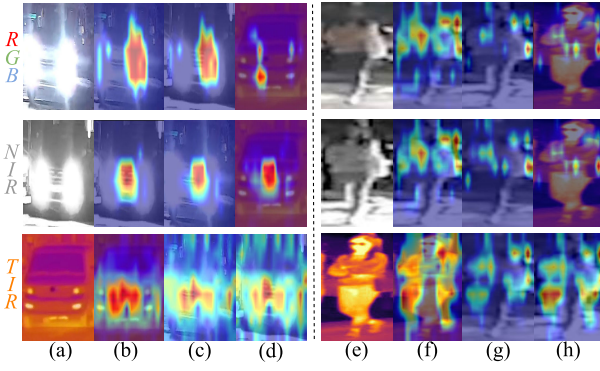


Fig. 11. The visualized attention maps. (a)–(d) show the results on the multi-modal vehicle dataset, including the input modality, the ground-truth activation map, and the reconstructed results of the other two missing modalities when the current modality is used as the available modality. (e)–(h) show the corresponding results on the multi-modal person dataset.

available modalities fail to provide sufficient identity cues, thus affecting the reconstruction of missing modalities. These results indicate that although REMIND shows strong robustness under modality-missing scenarios, its performance is still limited when the available modality is severely degraded. Developing more robust methods for modality-missing Re-ID under complex degradation conditions remains an important direction for future work.

V. CONCLUSION

This paper proposes a novel framework, Retrieval-Augmented Reconstruction with Dual Memories (REMIND), to address the challenging modality-missing object Re-ID task. Specifically, the dual-memory construction module decouples modality-specific features from modality-common features and stores them in a structured memory bank to enable effective reconstruction guided by priors. Additionally, the retrieval-augmented missing reconstruction module generates diverse reconstruction candidates using controlled perturbations and a lightweight ReBlock, thereby improving the model's robustness and discriminative ability. Extensive experiments conducted on multiple multi-modal Re-ID benchmark datasets demonstrate that the REMIND framework significantly outperforms existing state-of-the-art methods.

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